A covariance proof and a boundary counterexample for two digamma conjectures

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Abstract

We resolve both assertions of Conjecture 1 in the paper of Yang, Chu and Zhang on complete monotonicity and the digamma function. The ratio in part (i) is strictly increasing, by a Stieltjes integral representation and the strict covariance inequality. Part (ii) is false: at the proposed boundary parameter 155/294, the asserted completely monotone function has second derivative tending to minus infinity at zero. The proposed parameter range does characterize its ordinary nonnegativity.

Task 1: Resolve MathDB problem 328867

Let

$$R(x) = \psi(x + 1/2) - \log x, \qquad x > 0,$$

where $\psi = \Gamma'/\Gamma$, and put

$$G_a(x) = 24\left(x^4 + ax^2 + \frac{7}{40}a - \frac{31}{336}\right)R(x) - \left(x^2 - \frac{7}{40} + a\right), \tag{1}$$

$$L(x) = \frac{-24(x^4 - 31/336)R(x) + x^2 - 7/40}{24(x^2 + 7/40)R(x) - 1}. \tag{2}$$

These are the definitions in [1, Sections 3–4]; the same conjecture appears in [2, Conjecture 1, p. 14]. Part (i) asks whether L is increasing on $(0, \infty)$. Part (ii) asserts that $-G_a$ is completely monotone on $(0, \infty)$ if and only if $a \le 155/294$.

Theorem 1. *The denominator in* (2) *is strictly positive. The function L is strictly increasing and has exact range*

$$\frac{155}{294} < L(x) < \frac{11165}{8284} \qquad (x > 0).$$

Thus part (i) holds. Part (ii) is false: for $a_0 = 155/294$,

$$(-G_{a_0})''(x) = 48a_0 \log x + O(1) \longrightarrow -\infty \quad \text{as } x \downarrow 0.$$

In contrast, $-G_a(x) \ge 0$ for every $x > 0$ if and only if $a \le a_0$; in this range the pointwise inequality is always strict.

A positive Stieltjes representation

The differentiated Binet formula [3, Eq. 5.9.15] is

$$\psi(x) = \log x - \frac{1}{2x} - 2\int_0^\infty \frac{t}{(x^2+t^2)(e^{2\pi t}-1)}\,dt \qquad (x>0).$$

Apply this at x and $2x$, use $\psi(x+1/2) = 2\psi(2x) - \psi(x) - 2\log 2$, and change variables in the integral at $2x$. The identity

$$\frac{1}{e^u-1} - \frac{2}{e^{2u}-1} = \frac{1}{e^u+1}$$

gives

$$R(x) = 2\int_0^\infty \frac{t}{(x^2+t^2)(e^{2\pi t}+1)}\,dt. \tag{3}$$

All integrals above converge absolutely for $x>0$.

Let U have probability density

$$\frac{24}{e^{2\pi\sqrt{u}}+1}, \qquad u>0. \tag{4}$$

For integers $k \ge 0$, substituting $u = t^2$ gives

$$\mathbb{E}U^k = 48\int_0^\infty \frac{t^{2k+1}}{e^{2\pi t}+1}\,dt = \frac{48\Gamma(2k+2)(1-2^{-2k-1})\zeta(2k+2)}{(2\pi)^{2k+2}}.$$

For example, this follows by expanding $1/(e^v+1)$ in exponentials; the sum of the absolute term integrals is finite for $2k+2>1$. In particular,

$$\begin{gathered}\mathbb{E}1 = 1, \qquad m := \mathbb{E}U = \frac{7}{40}, \qquad b := \mathbb{E}U^2 = \frac{31}{336},\\ c := \mathbb{E}U^3 = \frac{127}{1280}, \qquad v := \mathrm{Var}(U) = b - m^2 = \frac{2071}{33600} > 0.\end{gathered} \tag{5}$$

The density has all moments and is positive throughout $(0,\infty)$. For $s>0$, define

$$S(s) = \mathbb{E}\frac{1}{s+U} = 24R(\sqrt{s}), \qquad D(s) = (s+m)S(s) - 1.$$

Strict Jensen gives $S(s) > 1/(s+m)$, hence $D(s)>0$. This proves positivity of the denominator of L.

The covariance inequality proves strict monotonicity

In terms of $s = x^2$, equation (2) becomes

$$\ell(s) := L(\sqrt{s}) = \frac{(b-s^2)S(s) + s - m}{D(s)} = m - s + v\frac{S(s)}{D(s)}. \tag{6}$$

Write $V_s = (s+U)^{-1}$. Differentiation under the expectation is permitted locally for $s>0$, since its derivatives are bounded there. Thus $S'(s) = -\mathbb{E}V_s^2$ and

$$\mathrm{Cov}(U, V_s) = \mathbb{E}\frac{U}{s+U} - mS(s) = 1 - (s+m)S(s) = -D(s).$$

Differentiating (6) now gives the exact identity

$$\ell'(s) = \frac{v(\mathbb{E}V_s^2 - S(s)^2) - D(s)^2}{D(s)^2} = \frac{\operatorname{Var}(U)\operatorname{Var}(V_s) - \operatorname{Cov}(U, V_s)^2}{D(s)^2} > 0. \tag{7}$$

The nonnegative sign is Cauchy–Schwarz for the centered random variables U and V_s. Equality would require V_s to be affine in U almost surely. The function $u \mapsto (s+u)^{-1}$ cannot agree with an affine function on an interval, whereas the density (4) is positive on every interval in $(0, \infty)$. The inequality is therefore strict. Since $L'(x) = 2x\ell'(x^2)$, part (i) follows on the entire positive half-line.

For completeness, the endpoint values can be checked without an uncontrolled asymptotic series. At zero, $R(x) = -\log x + O(1)$, so (2) gives

$$L(0+) = b/m = 155/294.$$

At infinity, finite geometric expansion of $(s+U)^{-1}$ gives

$$S(s) = s^{-1} - ms^{-2} + bs^{-3} - cs^{-4} + O(s^{-5}),$$

because the fourth moment is finite. Substitution in (6) yields

$$L(\infty) = \frac{c - mb}{b - m^2} = \frac{11165}{8284}.$$

Together with strict monotonicity these limits prove the asserted range.

A boundary counterexample to complete monotonicity

Set $a_0 = b/m = 155/294$. Since $ma_0 - b = 0$, equation (1) gives

$$-G_{a_0}(x) = a_0 - m + x^2 - 24x^2(x^2 + a_0)R(x). \tag{8}$$

The function $r(x) = \psi(x + 1/2)$ is analytic near zero and $R(x) = -\log x + r(x)$. Inserting this exact decomposition gives

$$-G_{a_0}(x) = a_0 - m + x^2 + 24x^2(x^2 + a_0)\log x - 24x^2(x^2 + a_0)r(x).$$

Twice differentiating this expression is legitimate for $x > 0$. The term $24a_0x^2 \log x$ contributes $48a_0 \log x + 72a_0$; the second derivative of $24x^4 \log x$ tends to zero; and the remaining analytic terms have bounded second derivatives at zero. Consequently,

$$(-G_{a_0})''(x) = 48a_0 \log x + O(1) \to -\infty.$$

Every completely monotone function has nonnegative second derivative at every positive point. Thus $-G_{a_0}$ is not completely monotone, even though a_0 is expressly included in the proposed parameter range. This refutes part (ii) analytically, without relying on a numerical test.

Finally, direct algebra gives

$$-G_a(x) = D(x^2)(L(x) - a).$$

As $D > 0$ and L has infimum a_0, this product is nonnegative for all $x > 0$ exactly when $a \le a_0$; strict monotonicity makes it positive even at $a = a_0$. The boundary counterexample therefore distinguishes ordinary positivity from the stronger derivative conditions asserted in the conjecture. This completes both parts of Theorem 1.

Task 2: Determine the actual complete-monotonicity cutoff

Find the sharp parameter a_* for which $-G_a$ is completely monotone exactly when $a \le a_*$. Using the source's Laplace kernel $Q(t) = t^{-1} - (2\sinh(t/2))^{-1}$, determine the infimum of

$$\frac{(31/336)Q(t) - Q^{(4)}(t)}{Q''(t) + (7/40)Q(t)}, \qquad t > 0,$$

and establish the number and location of its minimizing points.

Task 3: Derive higher-order bounds from the moment measure

Use the higher moments of (4) to construct successive rational upper and lower bounds for $R(x)$. Quantify their errors uniformly on positive intervals and compare the resulting bounds for harmonic numbers with existing DeTemple–Wang approximations.

Task 4: Classify the loss of convexity

Determine the number of positive zeros of $(-G_a)''$ as a varies between the actual complete-monotonicity cutoff and $155/294$. Describe their limiting behavior and identify the ranges of x where the conjectured functions retain ordinary convexity.

Task 5: Extend the covariance construction to shifted remainders

Construct analogous ratios for digamma remainders with shifts other than $1/2$. Specify the shifts and correction terms for which a positive Stieltjes measure exists, and derive the corresponding sharp endpoint constants and strict monotonicity statements.

References

[1] Z.-H. Yang, Y.-M. Chu and X.-H. Zhang, *Necessary and sufficient conditions for functions involving the psi function to be completely monotonic*, J. Inequal. Appl. **2015**, 157, Conjecture 1, Section 4. doi:10.1186/s13660-015-0674-8.

[2] Z.-H. Yang, *Several completely monotone functions related to DeTemple's sequence*, arXiv:1409.6411v1, 23 September 2014, Conjecture 1, page 14.

[3] NIST Digital Library of Mathematical Functions, Equation 5.9.15.